

NetComms 2 Summative 2 Mock Exam: Modules 3 and 4

50 items: choice questions plus dropdown matching. Answer key at the end.

Instructions: choose the best answer for every item. Matching items use a dropdown on the real exam, so pick one letter for each term. The answer key is at the end.

Part 1: Multiple Choice

1. Git and GitHub: which statement is correct?
 - A. They are the same thing
 - B. Git is a hosting service by Microsoft
 - C. GitHub is a repository hosting service that implements Git
 - D. GitHub is a distributed version control system

2. You need to build a request URI for a user resource. The scheme is https, the host is api.example.com, the port is 443, the path is /users/42, and the query string is ?verbose=true. Which part of the URI is the authority?
 - A. api.example.com:443
 - B. verbose=true
 - C. https
 - D. /users/42

3. When executing ``git pull``, which of the following occurs first?
 - A. A new merge commit is created on the local branch
 - B. All remote branches are deleted locally
 - C. The `.git` directory is updated with the latest commit and history
 - D. The working directory is updated with the latest content

4. What is the first step of the TDD cycle?
 - A. Deploy the code
 - B. Create a new test
 - C. Refactor the code
 - D. Write the application code

5. A SOAP message is returned with an error. Which element of the SOAP envelope contains the error details?
- A. <Fault>
 - B. <Envelope>
 - C. <Body>
 - D. <Header>
6. A client authenticates to a REST API using Basic authentication with the credentials alice:secret. Which exact header value should be included in the request?
- A. Bearer YWxpY2U6c2VjcmV0
 - B. Basic alice:secret
 - C. Authorization: Basic
 - D. Basic YWxpY2U6c2VjcmV0
7. Which files does PyTest automatically execute?
- A. Any file with .py
 - B. Scripts that start with test_ or end with _test.py
 - C. Only files named main.py
 - D. Files ending with _spec.py
8. Which rate limit algorithm processes queued requests at a fixed rate and rejects when the queue is full?
- A. Fixed window counter
 - B. Leaky bucket
 - C. Token bucket
 - D. Sliding window counter
9. Which is a purpose of rate limiting?
- A. Remove authentication
 - B. Cache responses
 - C. Protect against a Denial-of-Service (DoS) attack
 - D. Increase the response size

10. During a merge, Git cannot automatically combine changes and reports a merge conflict. Which of the following symbols in a .diff file indicates a line that was removed in one branch?
- A. /dev/null
 - B. -
 - C. @@
 - D. +
11. Your team uses Git for version control. You need to start work on a new feature without affecting the stable release branch. Which Git workflow best supports this need?
- A. Create a new feature branch from the release branch, develop, then merge back via a pull request
 - B. Commit directly to the main branch and push after testing
 - C. Delete the release branch and work on a temporary branch
 - D. Rebase the release branch onto your local master before starting work
12. You accidentally added a large confidential file to the staging area and realize you need to keep it in your working directory but stop tracking it in the repository. Which command should you use?
- A. `git rm --cached <file>`
 - B. `git rm <file>`
 - C. `git checkout -- <file>`
 - D. `git reset <file>`
13. While using Postman you receive a 429 Too Many Requests response. Which response header tells you exactly when the rate-limit window will reset?
- A. X-Rate Limit-Remaining
 - B. Retry-After
 - C. X-Rate Limit-Limit
 - D. X-Rate Limit-Reset
14. Which MVC component is the application's data structure that manages the data, logic, and rules?
- A. Model
 - B. View
 - C. Controller
 - D. Router

15. In a lab you need to call an HTTPS API that uses a self-signed certificate. Which parameter should you add to the Python requests call to ignore certificate verification?
- A. proxies
 - B. cert
 - C. verify=False
 - D. timeout
16. Which header does a request use to ask for compressed data?
- A. Content-Type
 - B. Accept-Encoding
 - C. Set-Cookie
 - D. Cache-Control
17. What must the subject in the Observer pattern be able to do?
- A. Poll each observer for changes
 - B. Store no data at all
 - C. Only send data once
 - D. Store a list of its observers and add or remove them
18. Which are the three most popular standard formats for exchanging information with remote APIs?
- A. CSV, XML, and SQL
 - B. HTML, CSS, and JS
 - C. XML, JSON, and YAML
 - D. JSON, YAML, and INI
19. What is one key difference between Git and other version control systems?
- A. Git stores data as snapshots instead of differences
 - B. Git stores only the delta between versions
 - C. Git works only on Linux
 - D. Git requires a lock before editing

20. Which status code means the client must first authenticate with a proxy?
- A. 409 Conflict
 - B. 403 Forbidden
 - C. 401 Unauthorized
 - D. 407 Proxy Authentication Required
21. A REST service returns JavaScript code as part of a response to extend client functionality. What is the main security concern with this code-on-demand feature?
- A. Inability to cache the response
 - B. Potential security vulnerabilities from executing third-party code
 - C. Higher network latency
 - D. Increased bandwidth usage
22. A developer wants to stage all files that have been modified, deleted, or added in the working directory for the next commit. Which Git command accomplishes this in a single step?
- A. `git commit -a`
 - B. `git push --all`
 - C. `git add .`
 - D. `git stage *`
23. You are editing a YAML configuration file that stores API keys. You need to add a comment explaining the purpose of each key. Where can you safely insert a comment?
- A. Only on lines that start with a dash '-'
 - B. At the beginning of the file, before any content
 - C. Inside a multiline string literal that holds a secret token
 - D. Anywhere in the file except inside a long string literal, preceded by '#'
24. After pushing his changes to the remote repository, Bob receives a conflict error because another teammate modified the same function. What does Git do to help resolve this situation?
- A. It creates a new branch named 'conflict-resolution' for manual merging
 - B. It stops the push and marks the conflicting files as needing manual resolution
 - C. It rewrites the commit history to hide the conflict
 - D. It automatically discards Bob's changes and keeps the remote version

25. In a YAML file that will be converted to JSON for an API payload, you need to ensure the value 12345 is treated as a string, not a number. Which tag should you use?
- A. !!float 12345
 - B. !!bool 12345
 - C. !!int 12345
 - D. !!str 12345
26. An API uses a sliding-window counter that permits 100 requests per 60-second window. You send 60 requests at time $t=0$, wait 30 seconds, then send another 50 requests. Will the 50th request of the second batch be accepted?
- A. Yes, because only 90 requests are in the current 60-second window
 - B. Yes, because the sliding window resets after each request
 - C. No, because the sliding window still includes the first 60 requests
 - D. No, because the limit applies to a fixed window, not a sliding one
27. Which REST constraint says the server cannot contain session states?
- A. Layered system
 - B. Stateless
 - C. Cache
 - D. Code-on-demand
28. A developer just finished implementing a new feature and ran all unit tests, which all passed. Before merging the changes, the team follows a process where a reviewer watches the developer type the final commit and provides feedback on the spot. Which type of code review is this?
- A. Email pass-around
 - B. Change-based code review
 - C. Over-the-shoulder code review
 - D. Formal code review
29. You are configuring a webhook that will notify your application whenever a new book is added. Which of the following is a mandatory requirement for the consuming application?
- A. The provider must send data in XML format
 - B. The client must register a URI with the provider
 - C. The client must use GET requests to receive notifications
 - D. The server must poll the webhook provider

30. Which is NOT a benefit of version control in the module?
- A. Enables collaboration
 - B. Faster hardware performance
 - C. Accountability and visibility
 - D. Work in isolation
31. Can server-side issues be resolved from the API client side?
- A. Yes, by changing the method
 - B. Yes, by changing the URI
 - C. No, contact the administrator of the API server
 - D. Yes, by turning off verify
32. You are writing unit tests for a Python module using the unittest framework. Which of the following is required to correctly define a test case class?
- A. Create any class and name test methods with 'check_'
 - B. Use only functions named 'unit_test' without a class
 - C. Inherit from unittest.TestCase and prefix test methods with 'test_'
 - D. Inherit from pytest.TestCase and use 'assert' statements
33. Your project uses a main branch called `master`. You need to start work on a new feature, then delete the feature branch after merging. Which sequence of commands is correct?
- A. `git branch -d feature && git checkout -b feature master && git merge feature`
 - B. `git checkout master && git branch -d feature && git merge feature`
 - C. `git branch feature && git checkout master && git branch -d feature`
 - D. `git checkout -b feature master && git merge feature && git branch -d feature`
34. Which status code means the request conflicts with the current state of the resource, for example an edit conflict?
- A. 409
 - B. 404
 - C. 415
 - D. 400

35. A team is adding a new form to an existing web application that follows the MVC pattern. The form collects user input and must validate it before updating the database. Which component should handle the validation logic?
- A. Repository, because it manages data persistence
 - B. Model, because it encapsulates the application's data and business rules
 - C. Controller, because it mediates between the view and the model
 - D. View, because it directly interacts with the user
36. A startup with three developers needs a version control system that allows them to work offline and later synchronize changes without a central server. Which type of VCS should they adopt?
- A. File-based manual backup system
 - B. Distributed Version Control System (DVCS)
 - C. Centralized Version Control System (CVCS)
 - D. Local Version Control System (LVCS)
37. Which statement about JSON is correct?
- A. JSON is a superset of YAML
 - B. JSON supports comments like XML
 - C. JSON does not support any standard way to include comments
 - D. Whitespace is significant in JSON
38. A development team notices that they often switch between fixing bugs, writing new features, and reviewing code, causing delays. Which of the seven Lean wastes does this situation illustrate?
- A. Partially done work
 - B. Motion
 - C. Task switching
 - D. Extra features
39. You run `git push origin develop` but receive an error about conflicts with the remote repository. What is the best immediate action to resolve this?
- A. Pull the remote changes, resolve any conflicts, then push
 - B. Force push with `git push -f`
 - C. Create a new branch and push there
 - D. Delete the remote branch and push again

40. Which API style is a request-response model that lets an application make a procedure call to another application?
- A. Remote Procedure Call (RPC)
 - B. GraphQL
 - C. Webhook
 - D. REST
41. A REST API call returns HTTP 400 with the message “No id field provided”. Which is the most likely cause of this error?
- A. You used the wrong HTTP method (e.g., GET instead of POST)
 - B. The server is currently overloaded
 - C. Your authentication token is invalid or expired
 - D. A required field is missing from the request body
42. What is YAML in relation to JSON?
- A. A superset of JSON designed for easier human readability
 - B. The parent of HTML
 - C. A subset of JSON
 - D. A binary version of JSON
43. You are building a real-time dashboard that displays live sensor data. Each widget on the dashboard must update automatically when the underlying sensor object changes. Which design pattern best fits this requirement and what is the key responsibility of the subject in this pattern?
- A. Singleton pattern; the subject ensures only one instance of the sensor exists
 - B. Observer pattern; the subject stores a list of observers and notifies them of changes
 - C. Decorator pattern; the subject adds new behavior to the sensor at runtime
 - D. Factory pattern; the subject creates widget objects on demand
44. Which REST constraint states responses must say whether they are cacheable?
- A. Stateless
 - B. Uniform interface
 - C. Cache
 - D. Client-server

45. Which response header specifies directives that must be obeyed by all caching mechanisms?

- A. Cache-Control
- B. Content-Type
- C. Accept-Encoding
- D. Set-Cookie

46. In Python, how are private variables and methods indicated by convention?

- A. An asterisk
- B. The word private
- C. A single leading underscore
- D. Double trailing underscore

Part 2: Matching

47. Match the Agile or Scrum term with its description.

- A. A list of resolved bugs
- B. Uses stand-up meetings to review progress
- C. A prioritized list of all the features for the software
- D. A simple statement of what a user needs and why
- E. A time-boxed period where working software is developed

Story: _____

Scrum team: _____

Sprint: _____

Backlog: _____

48. Match the MVC component with its role.

- A. The application's data structure; manages data, logic, and rules
- B. The visual representation of the data
- C. The middleman that takes user input and formats it for the model or view
- D. Stores a list of observers

Model: _____

View: _____

Controller: _____

49. Match the RESTful API method with its CRUD function.

- A. UPDATE
 - B. ERASE
 - C. CREATE
 - D. DELETE
 - E. READ
- GET: _____
- POST: _____
- DELETE: _____
- PUT/PATCH: _____

50. Match the rate limit algorithm with its description.

- A. Requests are compressed
 - B. A counter is assigned to a fixed window of time
 - C. Requests enter a queue and are processed at a fixed rate
 - D. Counts requests from the beginning of the window to the current time
 - E. Each user gets a defined number of tokens per time increment
- Leaky bucket: _____
- Token bucket: _____
- Fixed window counter: _____
- Sliding window counter: _____

Answer Key

No.	Answer	Why
1	C. GitHub is a repository hosting service that implements Git	GitHub slide
2	A. api.example.com:443	Scenario question; matches the slide wording
3	C. The .git directory is updated with the latest commit and history	Scenario question; matches the slide wording
4	B. Create a new test	TDD slide
5	A. <Fault>	Scenario question; matches the slide wording
6	D. Basic YWxpY2U6c2VjcmV0	Scenario question; matches the slide wording
7	B. Scripts that start with test_ or end with _test.py	Unit Testing slide
8	B. Leaky bucket	Leaky bucket
9	C. Protect against a Denial-of-Service (DoS) attack	What are Rate Limits?
10	B. -	Scenario question; matches the slide wording
11	A. Create a new feature branch from the release branch, develop, then merge back via a pull request	Scenario question; matches the slide wording
12	A. git rm --cached <file>	Scenario question; matches the slide wording
13	D. X-Rate Limit-Reset	Scenario question; matches the slide wording
14	A. Model	MVC slide
15	C. verify=False	Scenario question; matches the slide wording
16	B. Accept-Encoding	Compressed response data
17	D. Store a list of its observers and add or remove them	Observer slide
18	C. XML, JSON, and YAML	Data Formats
19	A. Git stores data as snapshots instead of differences	Git slide
20	D. 407 Proxy Authentication Required	2xx and 4xx status codes
21	B. Potential security vulnerabilities from executing third-party code	Scenario question; matches the slide wording
22	C. git add .	Scenario question; matches the slide wording
23	D. Anywhere in the file except inside a long string literal, preceded by '#'	Scenario question; matches the slide wording
24	B. It stops the push and marks the conflicting files as needing manual resolution	Scenario question; matches the slide wording
25	D. !!str 12345	Scenario question; matches the slide wording
26	C. No, because the sliding window still includes the first 60 requests	Scenario question; matches the slide wording
27	B. Stateless	REST: stateless
28	C. Over-the-shoulder code review	Scenario question; matches the slide wording

29	B. The client must register a URI with the provider	Scenario question; matches the slide wording
30	B. Faster hardware performance	Version Control Systems
31	C. No, contact the administrator of the API server	No response from API server
32	C. Inherit from unittest.TestCase and prefix test methods with 'test_'	Scenario question; matches the slide wording
33	D. git checkout -b feature master && git merge feature && git branch -d feature	Scenario question; matches the slide wording
34	A. 409	2xx and 4xx status codes
35	C. Controller, because it mediates between the view and the model	Scenario question; matches the slide wording
36	B. Distributed Version Control System (DVCS)	Scenario question; matches the slide wording
37	C. JSON does not support any standard way to include comments	JSON slide
38	C. Task switching	Scenario question; matches the slide wording
39	A. Pull the remote changes, resolve any conflicts, then push	Scenario question; matches the slide wording
40	A. Remote Procedure Call (RPC)	RPC slide
41	D. A required field is missing from the request body	Scenario question; matches the slide wording
42	A. A superset of JSON designed for easier human readability	YAML slide
43	B. Observer pattern; the subject stores a list of observers and notifies them of changes	Scenario question; matches the slide wording
44	C. Cache	REST: cache
45	A. Cache-Control	Response headers
46	C. A single leading underscore	Classes slide
47	Story = D; Scrum team = B; Sprint = E; Backlog = C	Agile Methods slides
48	Model = A; View = B; Controller = C	MVC slide
49	GET = E; POST = C; DELETE = D; PUT/PATCH = A	HTTP method table
50	Leaky bucket = C; Token bucket = E; Fixed window counter = B; Sliding window counter = D	Rate Limit Algorithms slides